Threat Assessment Report

Executive summary for model robustness under adversarial attack

Model: 20260206_171419_resnet50_classifier.pt

Attack Type: FGSM
Images Evaluated: 10

Attack Success Rate	Original Confidence	Adversarial Confidence	Execution Time
90.0%	46.59%	50.56%	35.38s Accuracy drop: -3.97%

Threat Level Assessment

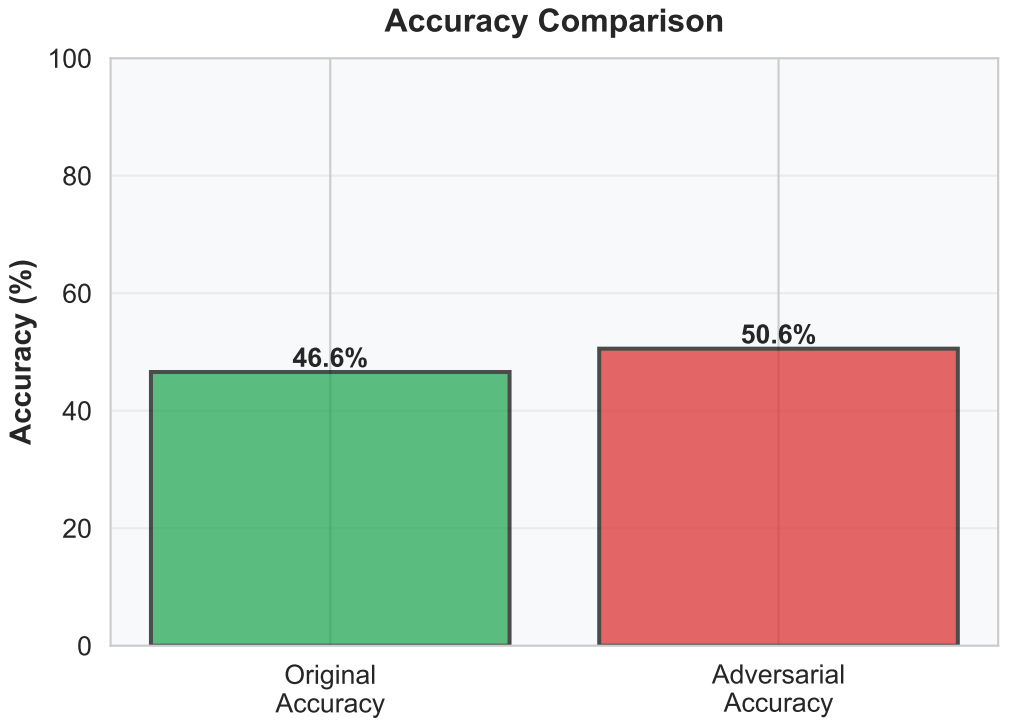
HIGH RISK

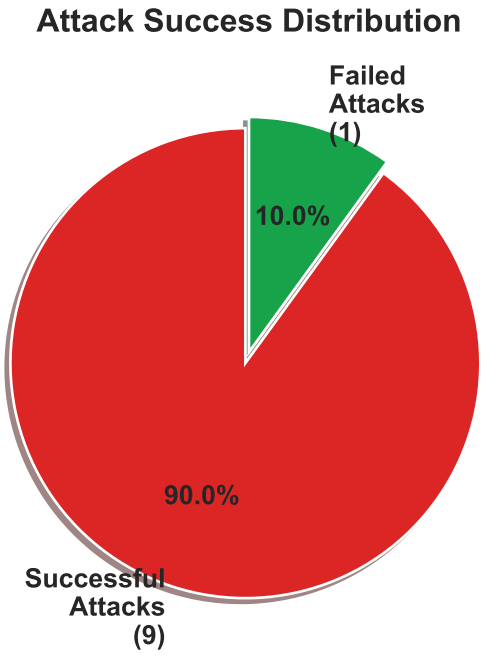
The FGSM attack altered model predictions on 90.0% of the evaluated images. Average confidence shifted from 46.59% to 50.56%, indicating the current robustness posture is high risk.

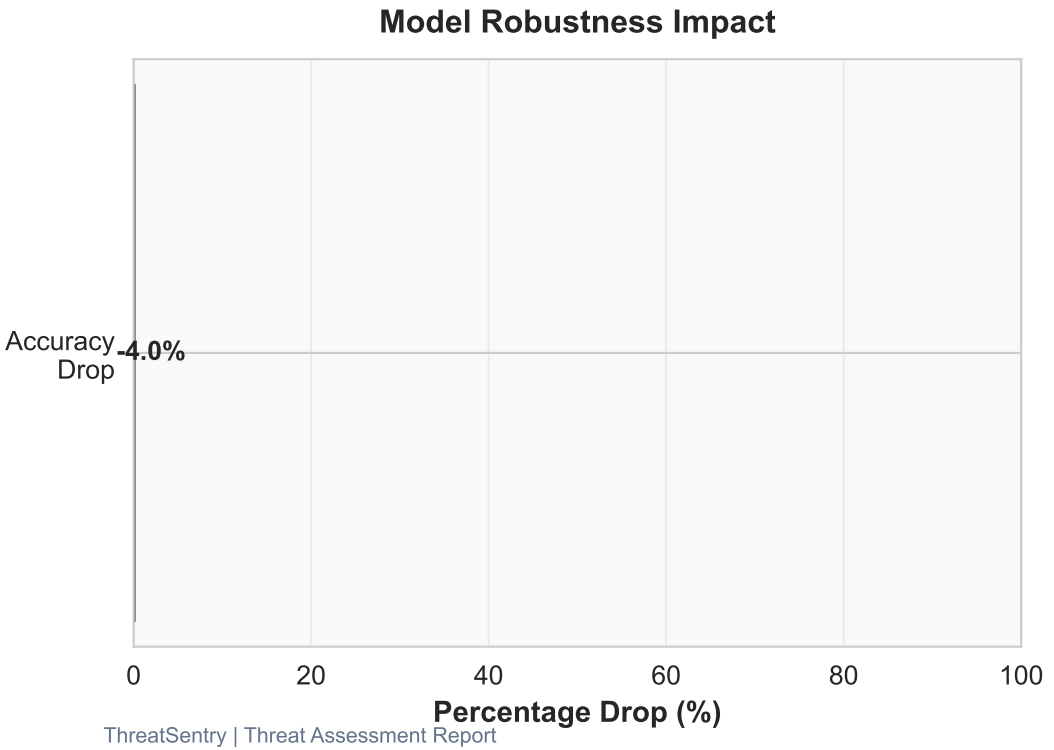
Assessment Narrative

Successfully executed FGSM attack on model 20260206_171419_resnet50_classifier.pt using 10 test images. Attack success rate: 90.0%. Average original accuracy: 46.59%, Average adversarial accuracy: 50.56%. The attack successfully fooled the model in 9 out of 10 cases.

Detailed Analysis







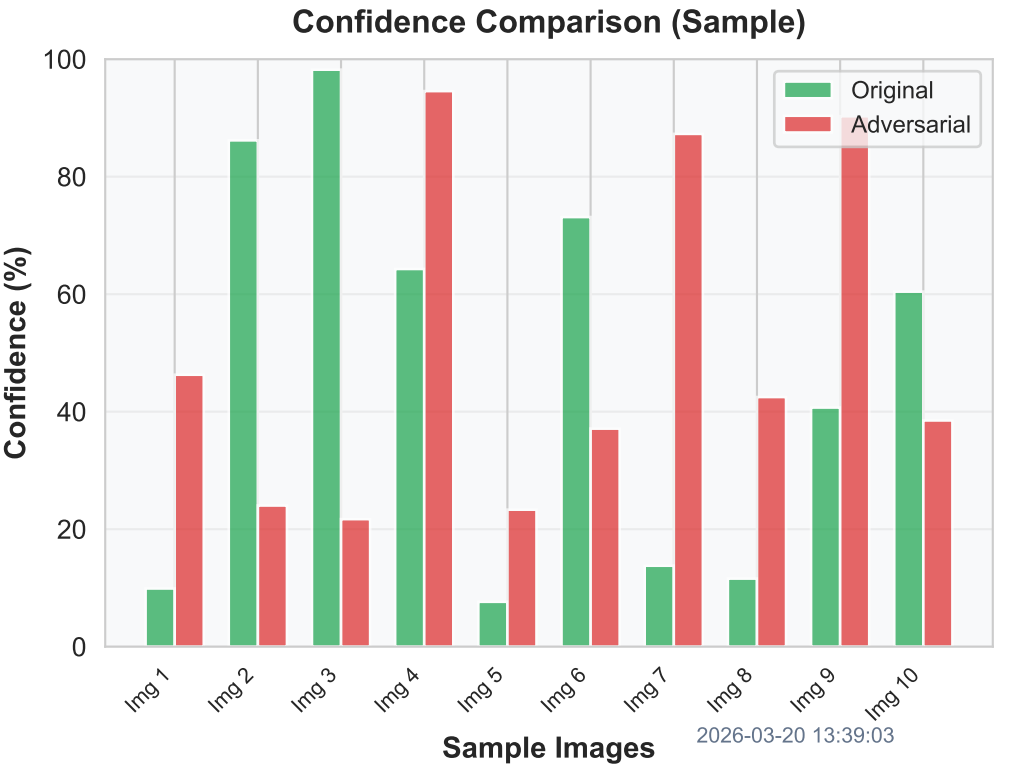
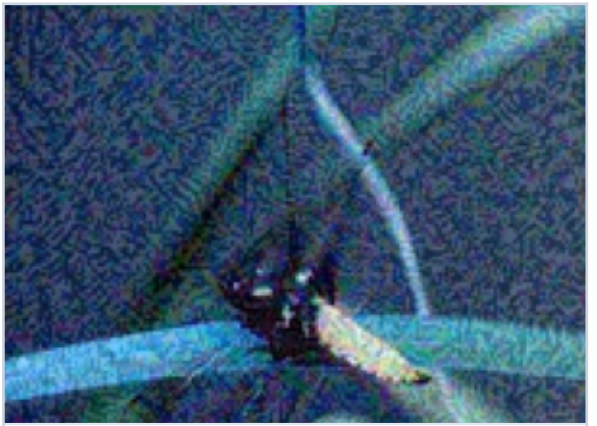


Image Evidence and Predictions

Images 1-4 of 10

Attack Successful
Original image: carousel.JPEG
Adversarial image: carousel.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 575
Original confidence: 40.71%
After attack predicted as: Class 611
Adversarial confidence: 90.29%

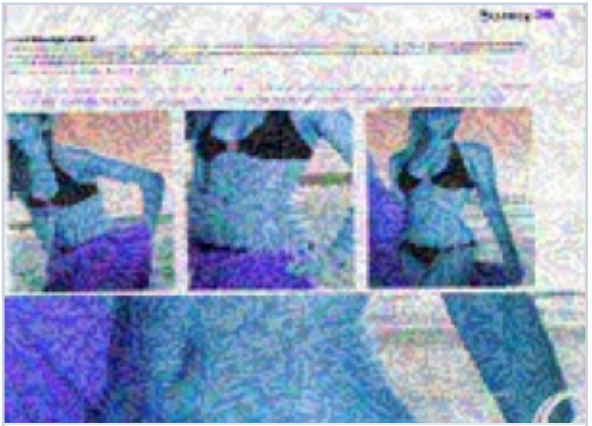


Attack Successful
Original image: dragonfly.JPEG
Adversarial image: dragonfly.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 542
Original confidence: 13.78%
After attack predicted as: Class 815
Adversarial confidence: 87.29%

Attack Successful
Original image: pitcher.JPEG
Adversarial image: pitcher.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 968
Original confidence: 7.64%
After attack predicted as: Class 868
Adversarial confidence: 23.32%

Attack Blocked
Original image: analog_clock.JPEG
Adversarial image: analog_clock.JPEG (attacked)
Outcome: Attack Blocked
Model predicted as: Class 892
Original confidence: 64.29%
After attack predicted as: Class 892
Adversarial confidence: 94.57%

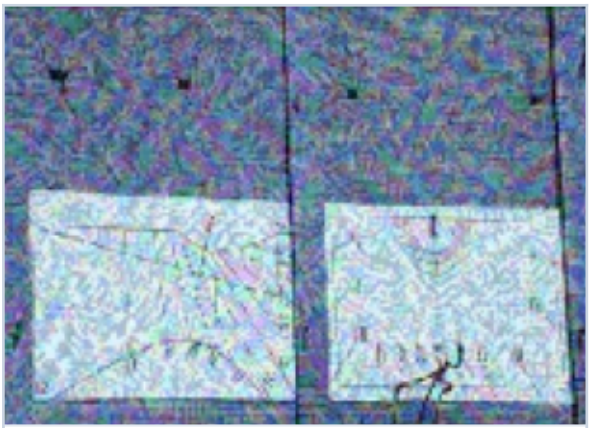
Image Evidence and Predictions



Attack Successful
Original image: bikini.JPEG
Adversarial image: bikini.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 916
Original confidence: 11.59%
After attack predicted as: Class 750
Adversarial confidence: 42.50%

Attack Successful
Original image: giant_schnauzer.JPEG
Adversarial image: giant_schnauzer.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 197
Original confidence: 86.19%
After attack predicted as: Class 206
Adversarial confidence: 24.03%

Attack Successful
Original image: African_grey.JPEG
Adversarial image: African_grey.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 87
Original confidence: 98.22%
After attack predicted as: Class 591
Adversarial confidence: 21.71%

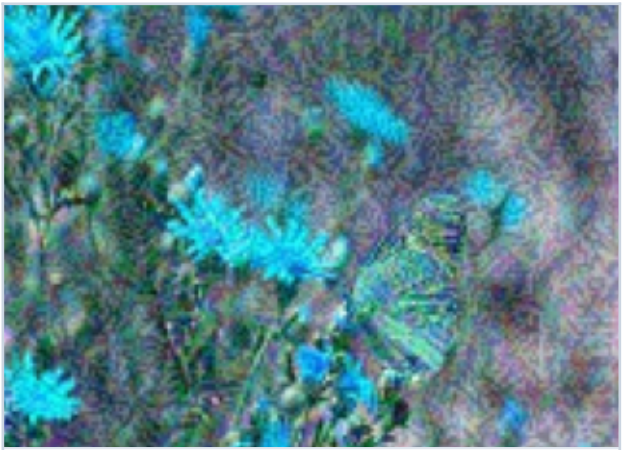


Attack Successful
Original image: sundial.JPEG
Adversarial image: sundial.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 646
Original confidence: 60.44%
After attack predicted as: Class 549
Adversarial confidence: 38.51%

Image Evidence and Predictions

Images 9-10 of 10

Attack Successful
Original image: dough.JPEG
Adversarial image: dough.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 618
Original confidence: 9.91%
After attack predicted as: Class 549
Adversarial confidence: 46.29%



Attack Successful
Original image: sulphur_butterfly.JPEG
Adversarial image: sulphur_butterfly.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 84
Original confidence: 73.13%
After attack predicted as: Class 794
Adversarial confidence: 37.12%

Recommendations

Security actions based on observed threat exposure

Assessment Summary

Successfully executed FGSM attack on model 20260206_171419_resnet50_classifier.pt using 10 test images. Attack success rate: 90.0%. Average original accuracy: 46.59%, Average adversarial accuracy: 50.56%. The attack successfully fooled the model in 9 out of 10 cases.

Security Recommendations

1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques